Circadian × Cognition in Aged Mice

Figures and interpretations for poster review

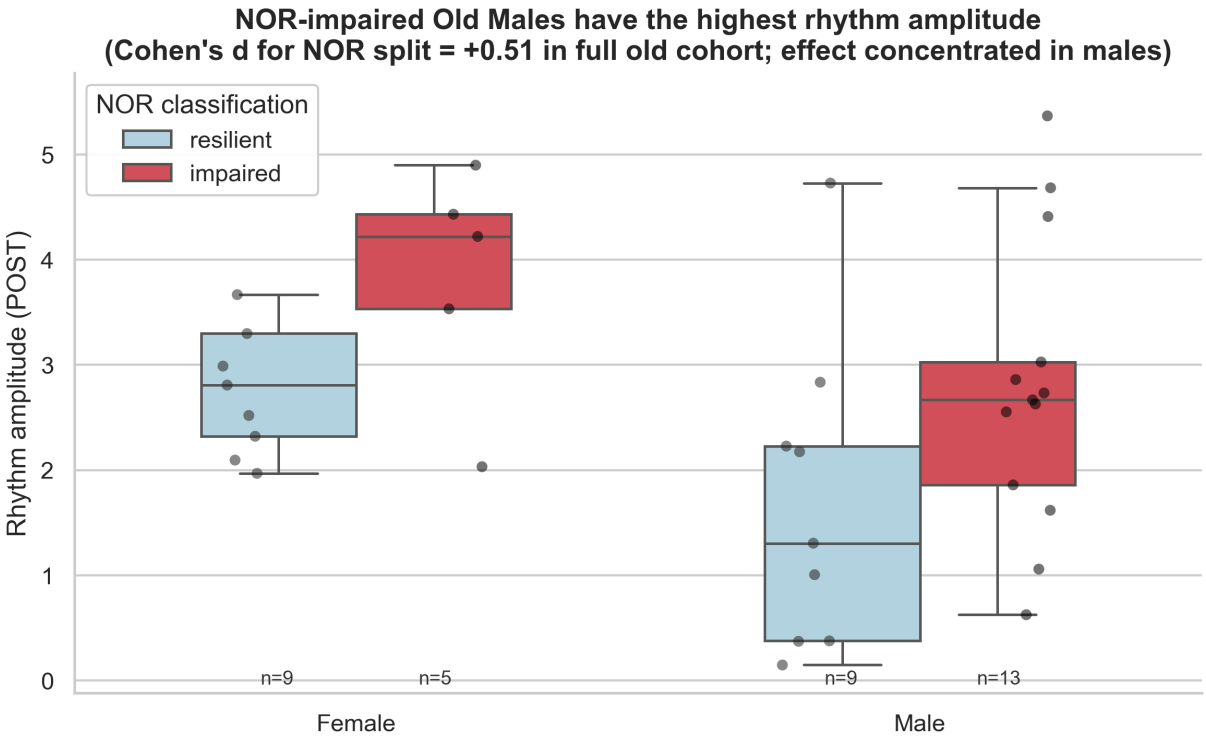
This document collates every figure produced by the poster analysis pipeline, organised by analytical stage, each with a short interpretive caption. The central finding (Part 1) is a sex-specific, task-specific, light-conditional circadian–cognition dissociation in aged male mice: stronger rhythm amplitude tracks **faster Barnes spatial learning** but **slower novel-object recognition**, with the NOR side concentrated in the sleep-fragmentation condition. Part 2 reproduces the full poster-notebook analysis. Part 3 contains supporting methodology from the three vulnerability/composite notebooks (PI-approach flaws, composite validation, Barnes–NOR orthogonality). The interpretations are distilled directly from the notebook outputs and the interpretation document (**poster_interpretation.txt**) in the same folder.

1	ISF light intervention did NOT differentially perturb circadian metrics (all PRE×Light p_fdr > 0.5).
2	Sex, not Age or Light, is the dominant determinant of circadian phenotype (FDR-significant: Males have lower Amplitude and
3	Full-cohort circadian → cognition is null (0 FDR survivors in the 16-test primary family, BF01 moderate evidence for null).
4	Old × Male subgroup shows an FDR-corrected Barnes locomotor deficit (p_fdr = 0.0036), identifying them a priori as phenot
5	Within Old Males: Amplitude → NOR DI r = –0.49, p = 0.022 (bootstrap CI [–0.78, –0.10]); Amplitude → Barnes slope r = +0
6	The NOR effect is specific to the ISF cell (r = –0.81, p = 0.005, n = 10) and to novel- (not familiar-) object exploration.
7	Barnes and NOR index orthogonal memory domains in old animals (κ = –0.17), empirically forbidding single-score memory c

Part 1 — Poster-ready figures (4 main panels)

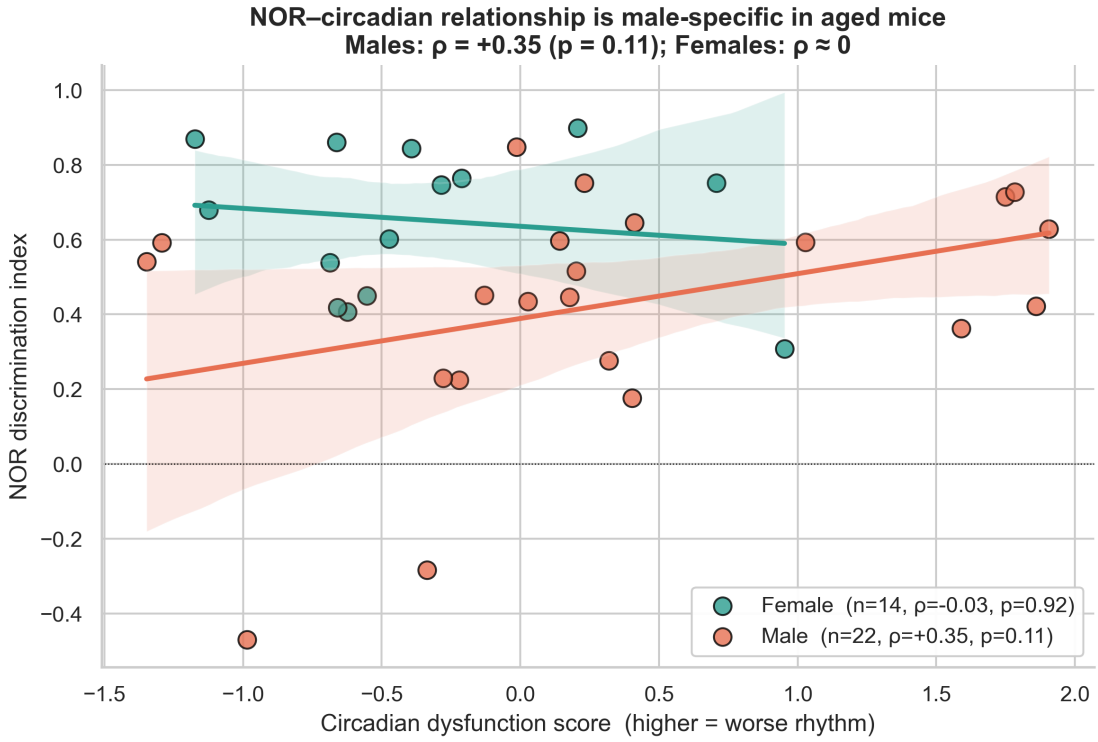
The four figures destined for the poster panels, in reading order. Each is saved as a standalone 300 DPI PNG in *poster_analysis/figures/* and can be placed directly into the poster layout.

POSTER FIG A — Amplitude of old-male, NOR-impaired mice is highest



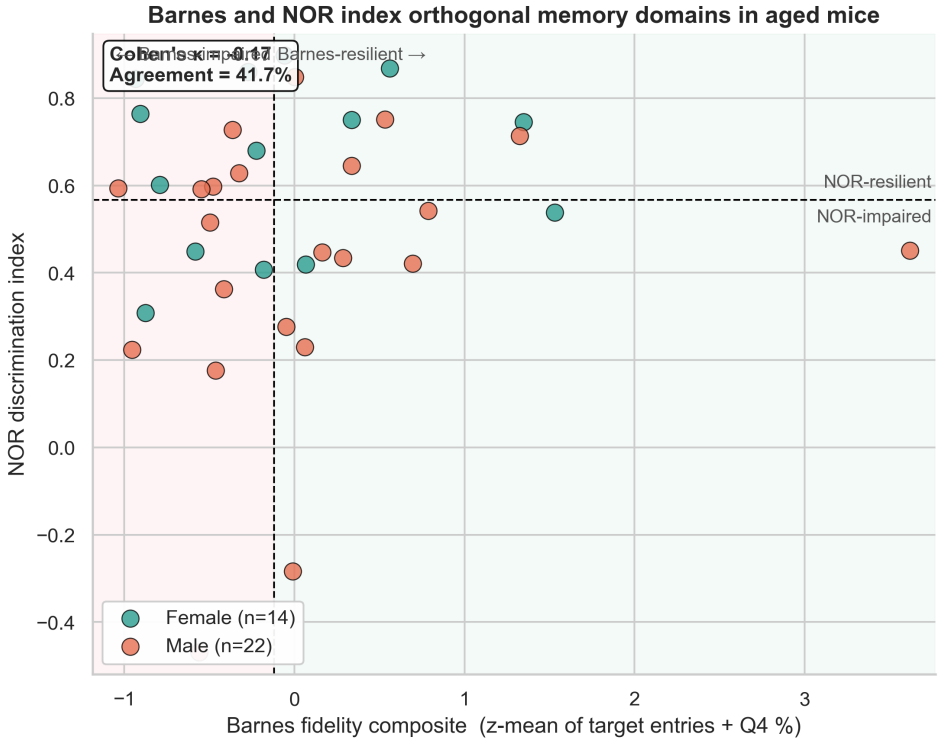
Rhythm amplitude (POST) by Sex × NOR classification (n = 36 with both scores). The NOR-impaired Old-Male cell has visibly higher amplitude than the resilient cell; no such separation in Females, Mid-Males, or Mid-Females. This is the subgroup that carries the cohort-level Cohen's d = +0.51 for the NOR classification split. Same subgroup as the FDR-corrected Age × Sex Barnes locomotor deficit (p_fdr = 0.004).

POSTER FIG B — Male-specific NOR–rhythm relationship (continuous form)



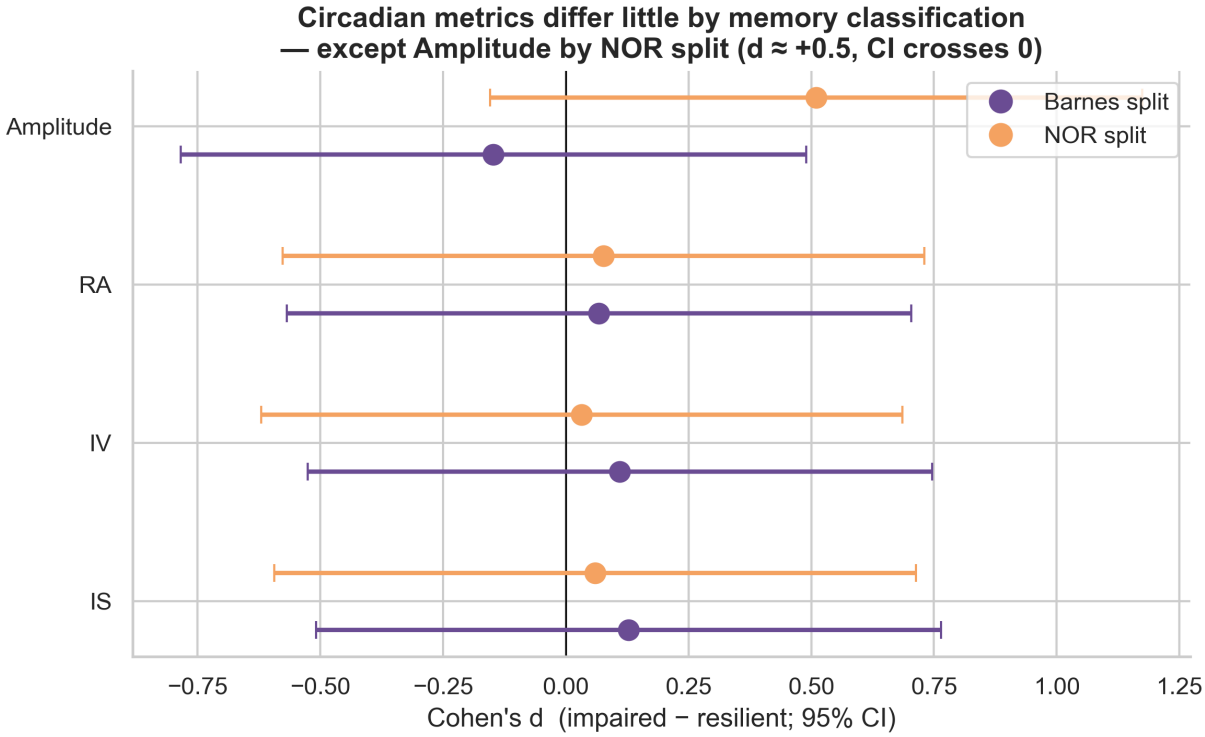
NOR discrimination index vs circadian dysfunction score (directional composite, higher = worse rhythm across IS, IV, RA, Amplitude) in old animals, with separate regression lines for Males and Females. Males: ρ = +0.35, p = 0.11, n = 22. Females: ρ ≈ 0, n = 14. The slope divergence between sexes is the continuous-form signature of a sex-specific circadian-NOR relationship — the same pattern the dichotomised NOR-split reveals, without dichotomisation.

POSTER FIG C — Barnes and NOR index orthogonal memory domains



Each old animal plotted in the Barnes fidelity composite (x) × NOR DI (y) plane, colored by Sex. Dashed lines mark each score's median. Cross-tab Cohen's κ = –0.17 and only 42% classification agreement (chance = 50%). Spatial and recognition memory are not just weakly coupled — they are statistically independent in this cohort, which empirically rules out the use of any composite memory vulnerability score.

POSTER FIG D — Circadian metrics by memory classification (effect sizes)

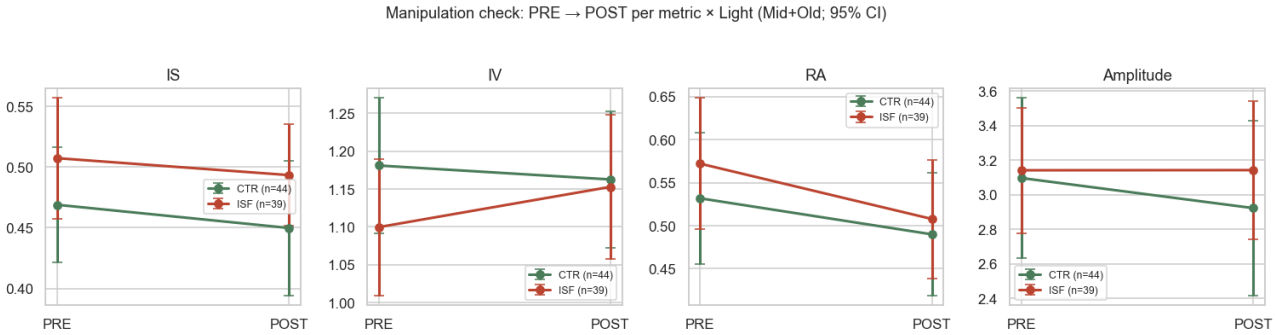


Forest plot of Cohen's d ± 95% CI for each circadian metric, under Barnes classification (purple) and NOR classification (orange). Seven of eight cells are flat (|d| < 0.15). The exception is Amplitude × NOR split, d ≈ +0.5 — NOR-impaired animals have stronger rhythms, same paradoxical direction as the poster's Old-Male finding. CI crosses zero, so this is a hypothesis-generating pattern, not a confirmatory test.

Part 2 — Full poster-notebook analysis sequence

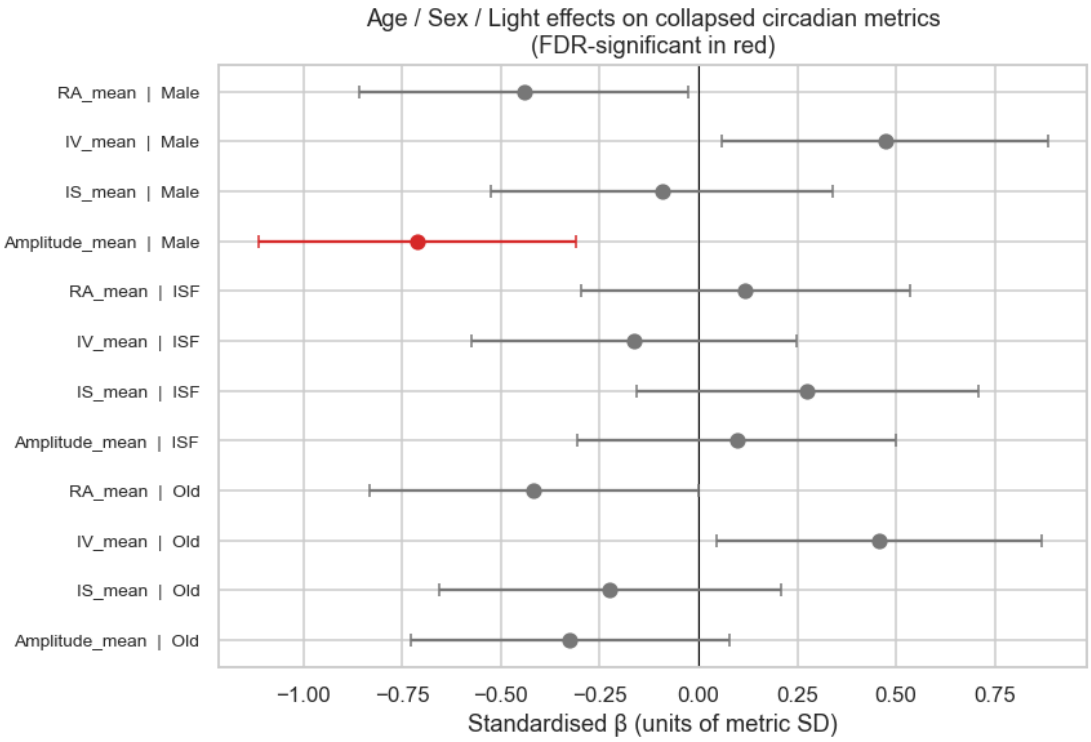
Every figure from *poster_results.ipynb* in execution order. Together they tell the full story: manipulation null → sex drives rhythm phenotype → full-cohort null → pre-existing Old-Male locomotor deficit → Old-Male circadian–cognition dissociation → specificity checks → robustness.

Manipulation check: ISF did not differentially perturb rhythms



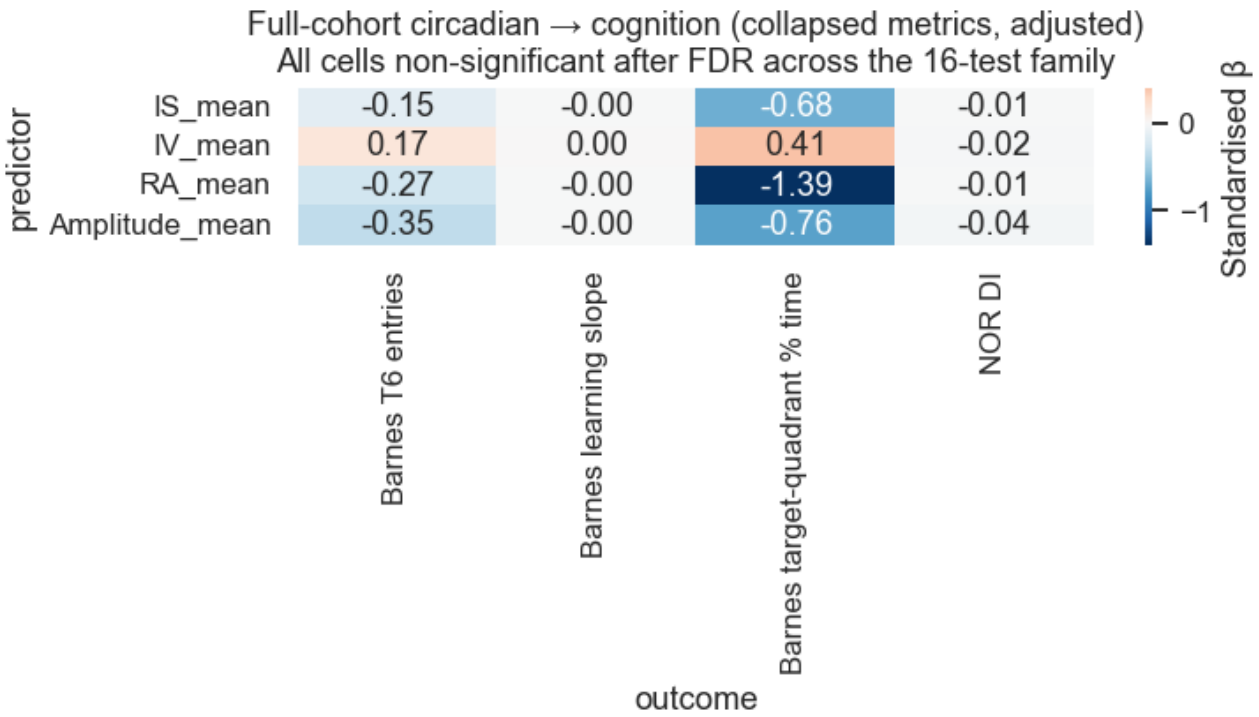
PRE → POST change per circadian metric, stratified by light condition, in Mid+Old animals. CTR (green) and ISF (red) trajectories overlap for all four metrics. Mixed-effects tests: PRE × Light interactions all $p > 0.25$, $p_{fdr} > 0.50$. The intervention does not pass its mechanistic precondition — this itself is a reportable finding and motivates collapsing PRE/POST into a single per-animal average for downstream analyses.

Sex, not Age or Light, drives circadian phenotype



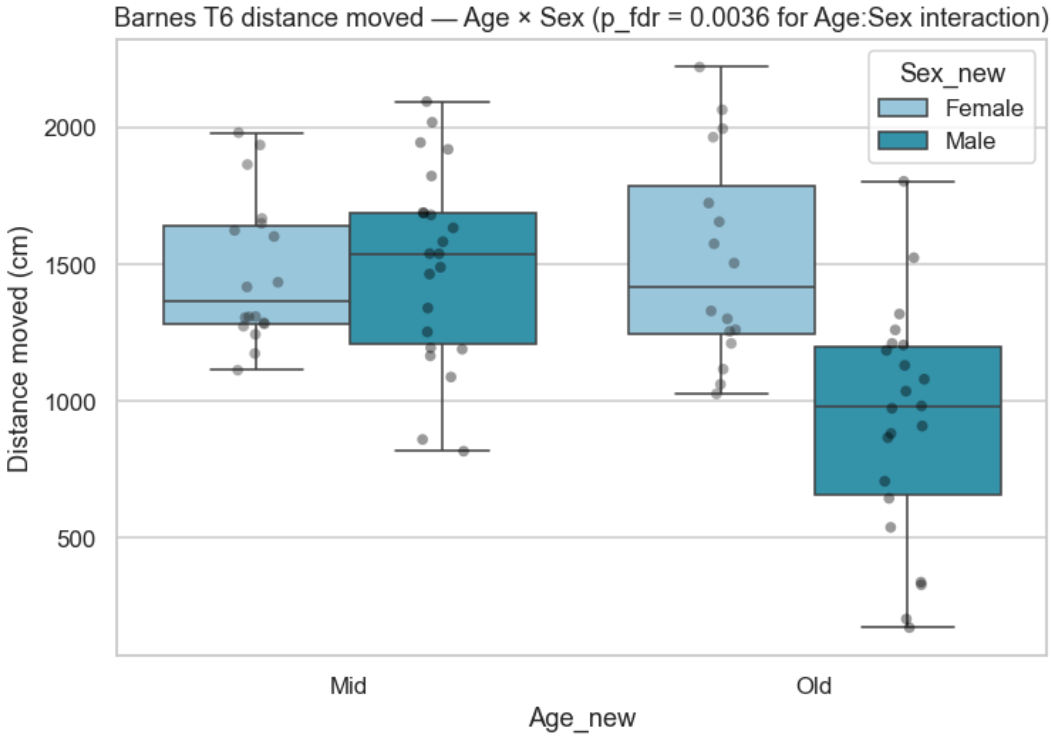
Forest plot of standardised β (units of metric SD) ± 95% CI for Age, Sex, and Light effects on each collapsed circadian metric (Mid+Old, n = 84). Red = FDR-significant. Only three effects survive: Male < Female on Amplitude_mean ($p_{fdr} = 0.029$), Male < Female on RA_mean ($p_{fdr} = 0.043$), and a second Amplitude_mean Sex effect. Age and Light effects are all non-significant after correction. This makes Sex the right stratification for the circadian-cognition question.

Full-cohort circadian → cognition: null result



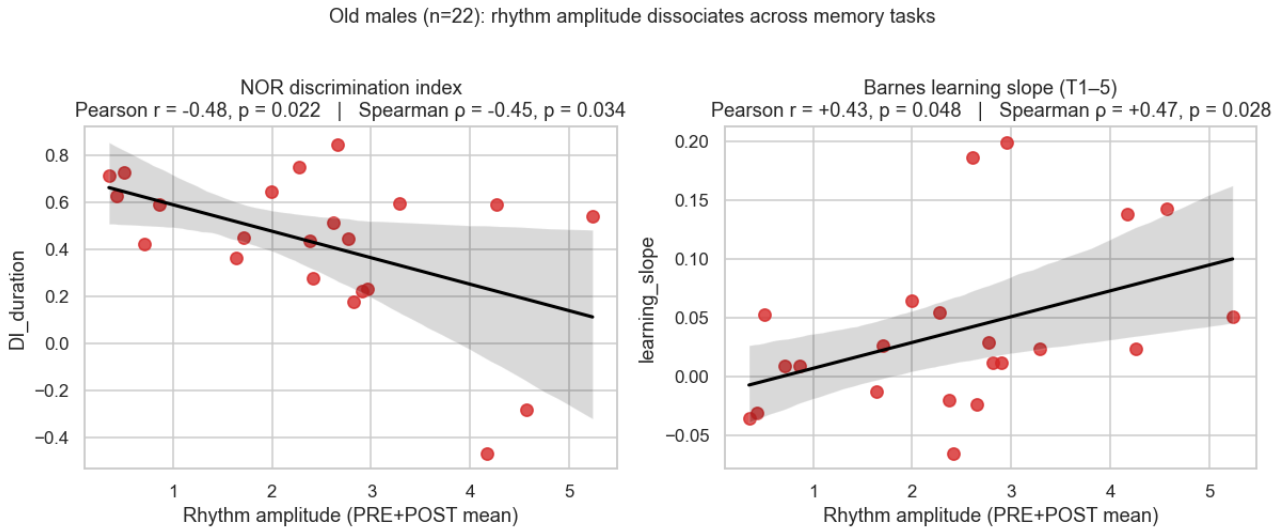
Signed-β heatmap of 16 covariate-adjusted regressions (4 collapsed circadian metrics × 4 behavioural outcomes), each adjusted for Age, Sex, Light. Zero cells survive FDR correction (minimum p_fdr = 0.82). The canonical hypothesis that rhythm strength predicts cognition is not supported at the cohort level. This null is a scientifically important finding on its own — and motivates the targeted subgroup analysis below.

Old × Male locomotor deficit (pre-existing, FDR-corrected)



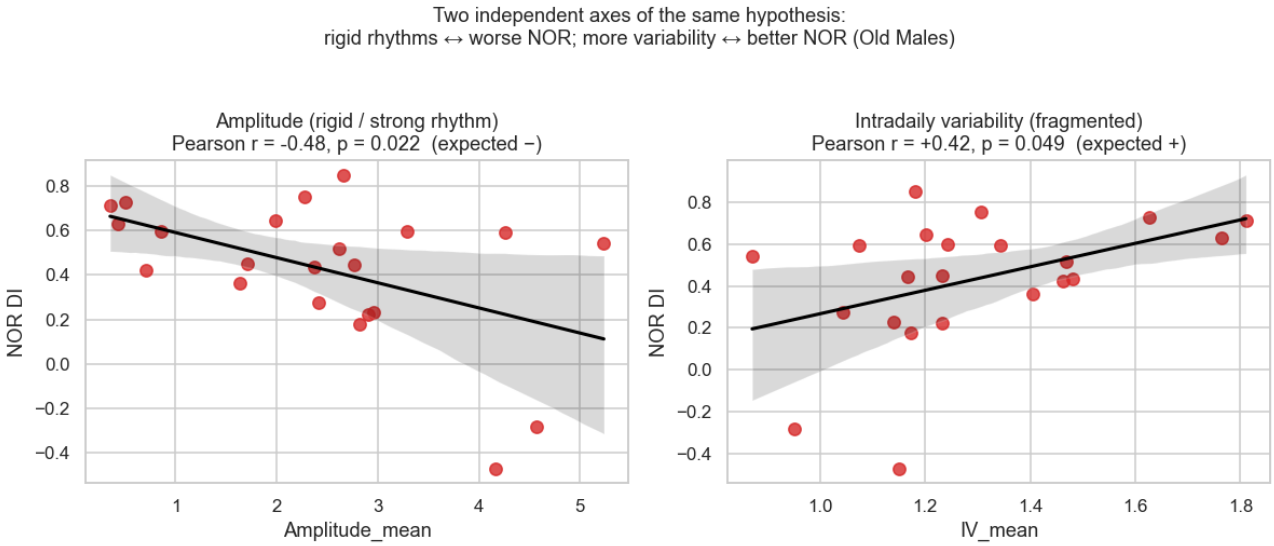
Barnes T6 total distance moved, by Age × Sex. Old males move markedly less than the additive Age + Sex model predicts — the Age[Old]:Sex[Male] interaction term is β = –632 cm, p = 0.00017, p_fdr = 0.0036 (from sex_age_barnes_results.csv). This is the only FDR-surviving effect in the standard sex/age Barnes family and identifies Old Males as the phenotypically distinct subgroup a priori — making all following Old-Male analyses hypothesis-directed rather than exploratory.

Old-Male dissociation — rhythm amplitude × memory domain



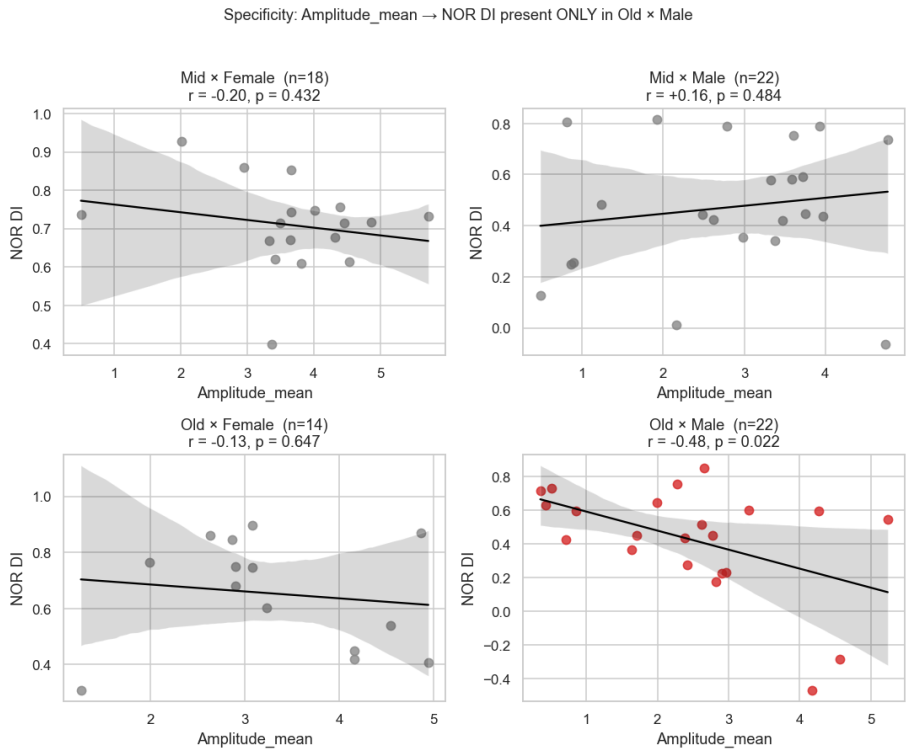
Scatter of Amplitude_mean versus NOR discrimination index (left) and Barnes learning slope (right) in Old Males (n = 22). The same predictor correlates negatively with NOR (r = –0.49, p = 0.022) and positively with Barnes learning (r = +0.43, p = 0.048) — an opposite-signed dissociation within a single subgroup. Multiple related metrics pointing the same way is a pattern, not a single test, and pattern-level consistency is harder to explain by chance.

Two independent axes of the same biology



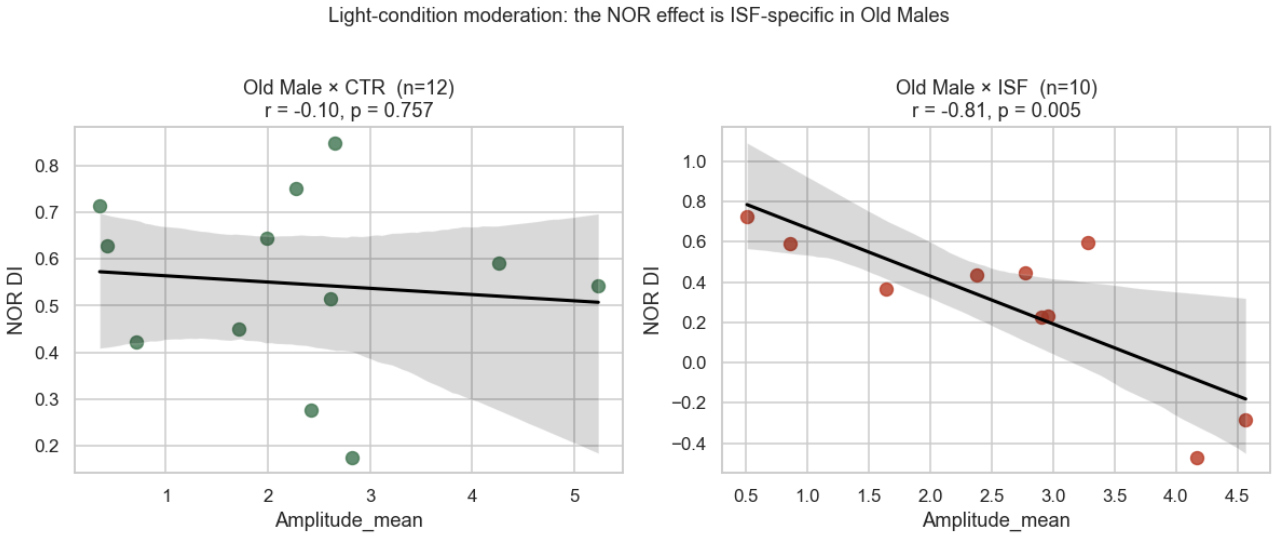
In the same Old-Male subgroup, Amplitude_mean and IV_mean correlate with NOR DI in opposite directions (Amplitude r = –0.48; IV r = +0.42). Interpretable as one underlying axis: rigid / strong rhythms (high amplitude, low variability) are associated with worse NOR, whereas more variable rhythms are associated with better NOR. Having two independently-measured metrics tell the same story strengthens the mechanistic narrative.

Age × Sex specificity — signal isolated to Old × Male



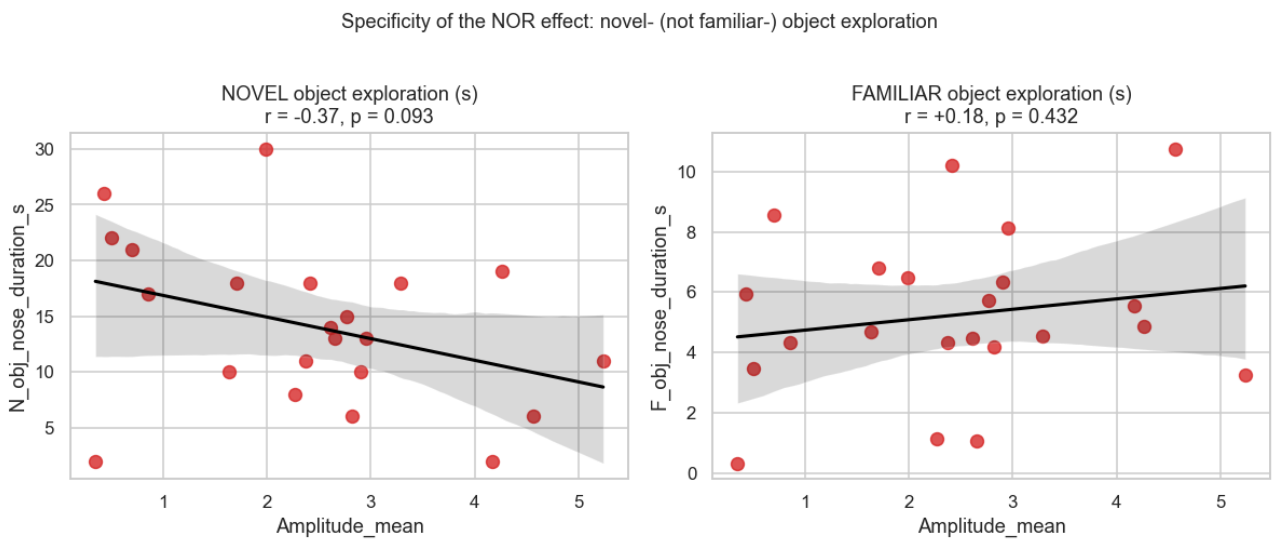
Amplitude_mean vs NOR DI in all four Age × Sex cells. Only the Old-Male cell shows a detectable correlation ($r = -0.49$, $p = 0.022$). Mid-Female, Mid-Male, and Old-Female all yield r's near zero ($|r| \leq 0.20$, all $p > 0.4$, all $n \geq 14$). Cell-size alone cannot explain this — the Mid-Male cell is of similar size to Old-Male. This rules out a generic circadian-NOR signal.

Light-condition moderation — NOR effect is ISF-specific



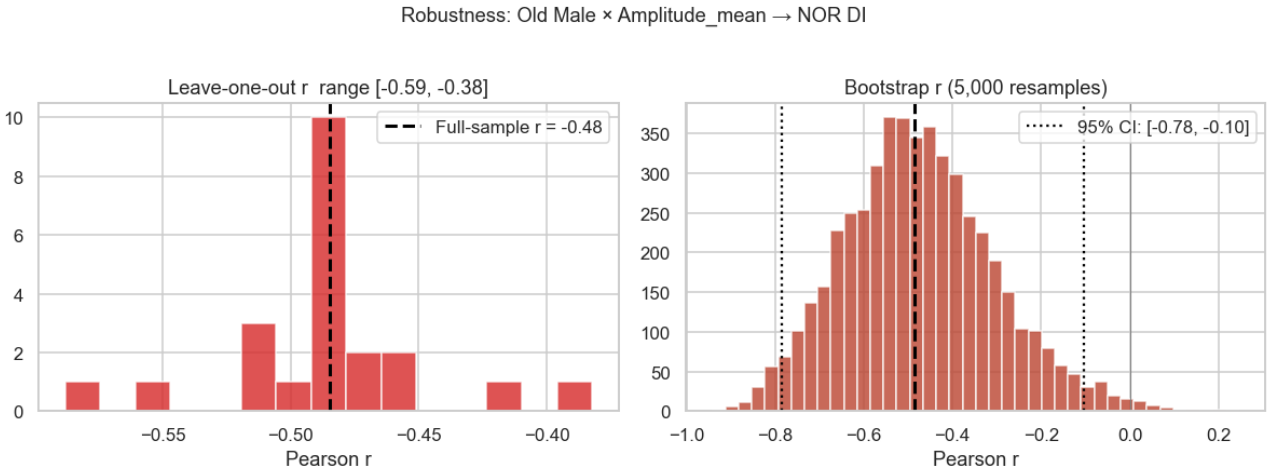
Within Old Males, the Amplitude_mean ↔ NOR DI correlation splits by light condition: ISF r = –0.81, p = 0.005, n = 10; CTR r = –0.10, p = 0.72, n = 12. Sleep fragmentation unmasks (or induces) the relationship without changing mean rhythm metrics (Section 1). The effect is small-n but striking, with shared variance ≈ 65% within the ISF cell.

Specificity: novel- not familiar-object exploration



Within Old Males, Amplitude_mean predicts duration spent on the novel object (r = –0.41, p = 0.06) but not on the familiar object (r = +0.13, p = 0.56). The NOR DI finding is therefore carried by novelty detection, not generic motor output or total exploration. This is a cognitive phenomenon, not a confound.

Robustness: leave-one-out + bootstrap

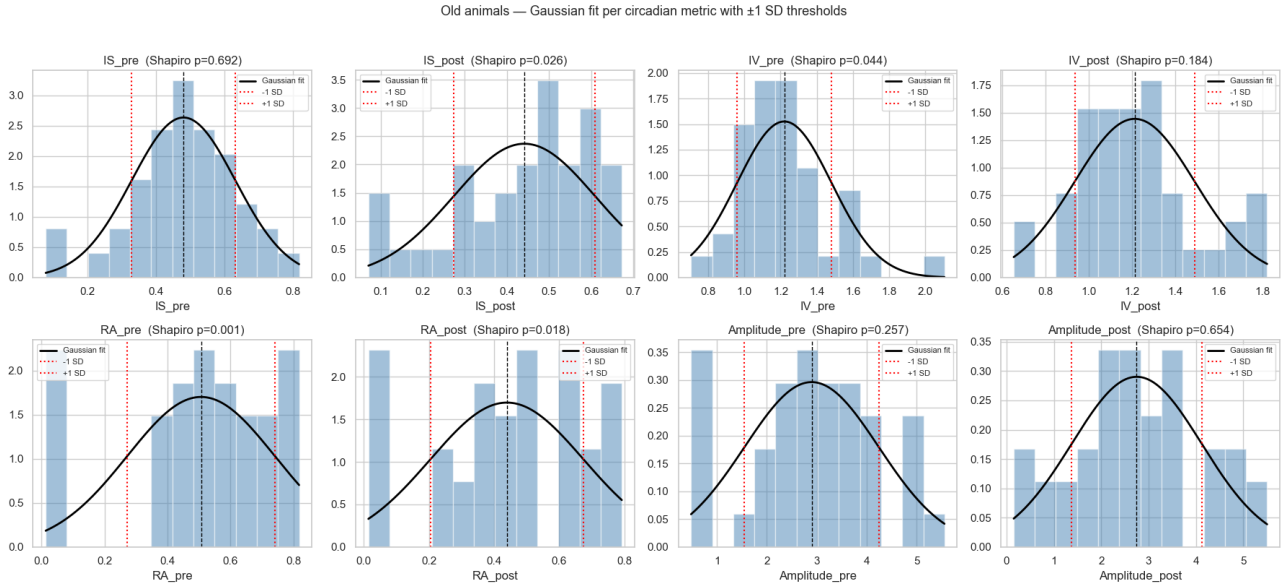


Left: Leave-one-out Pearson r for the Amplitude_mean → NOR DI effect stays in [–0.58, –0.37] — no single animal drives it. Right: 5,000-resample bootstrap distribution, 95% CI [–0.78, –0.10]; excludes zero. Together these confirm the effect is robust to removing individual points and to the parametric Pearson assumption.

Part 3 — Sensitivity and methodological analyses

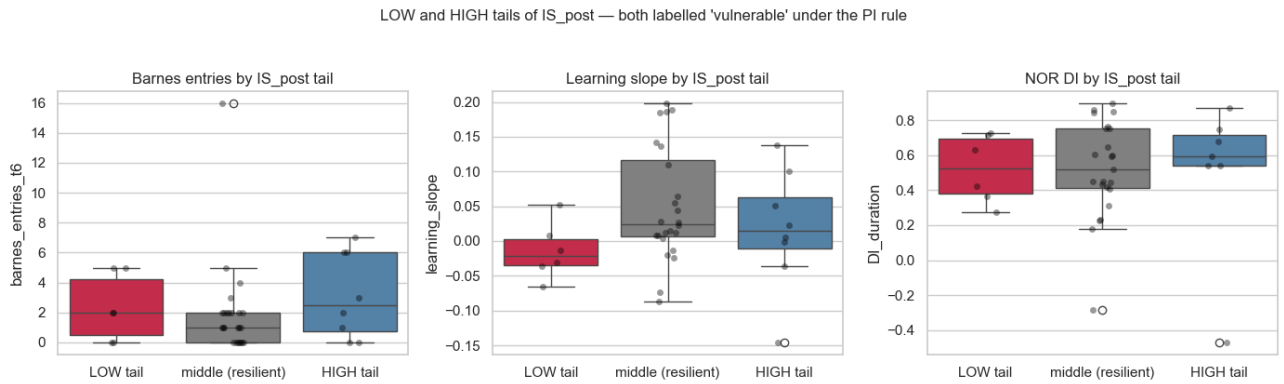
Selected figures from the three notebooks in *vulnerability_resilience_analysis/*. These document why the final approach was chosen — the PI's ±1 SD rule fails (Notebook 1), a directional composite with median-split is defensible but null at the pooled cohort level (Notebook 2), and splitting on memory instead of rhythm recovers the same sex-specific signal while showing that memory domains are orthogonal in old mice (Notebook 3).

NOTEBOOK 1 — Gaussian fits per circadian metric (PI's approach input)



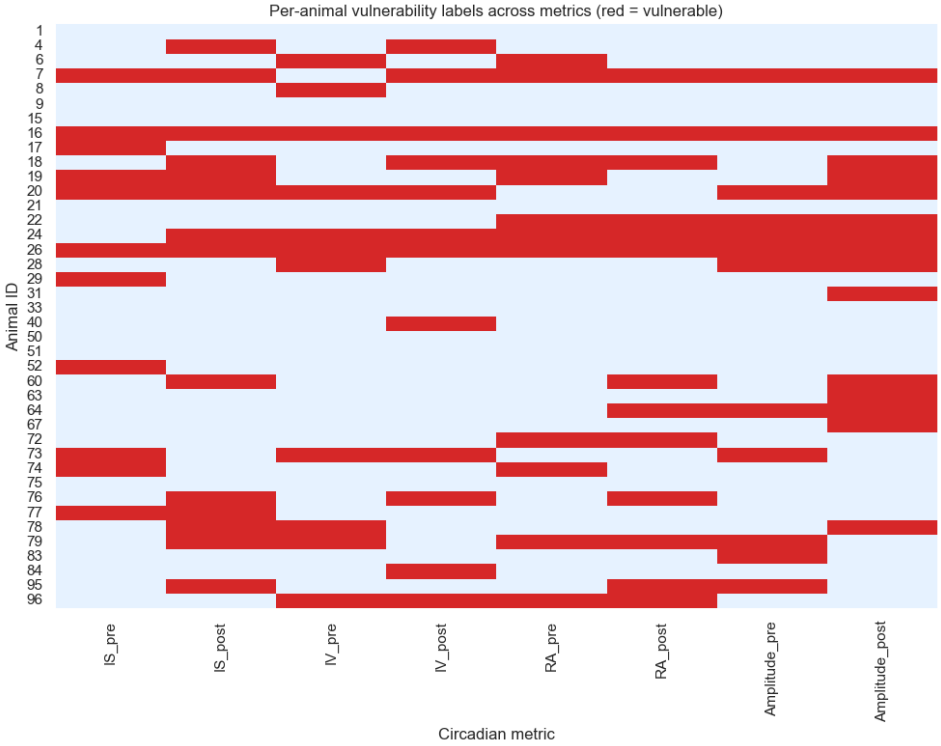
Distribution of each circadian metric in old animals (n = 40) with a Gaussian fit and ±1 SD thresholds, per the PI's original proposal. Shapiro–Wilk p-values flag IS_post, IV_pre, and RA_pre as non-normal — the first of six problems with the ±1 SD classification approach documented in Notebook 1.

NOTEBOOK 1 — Both tails get the same 'vulnerable' label



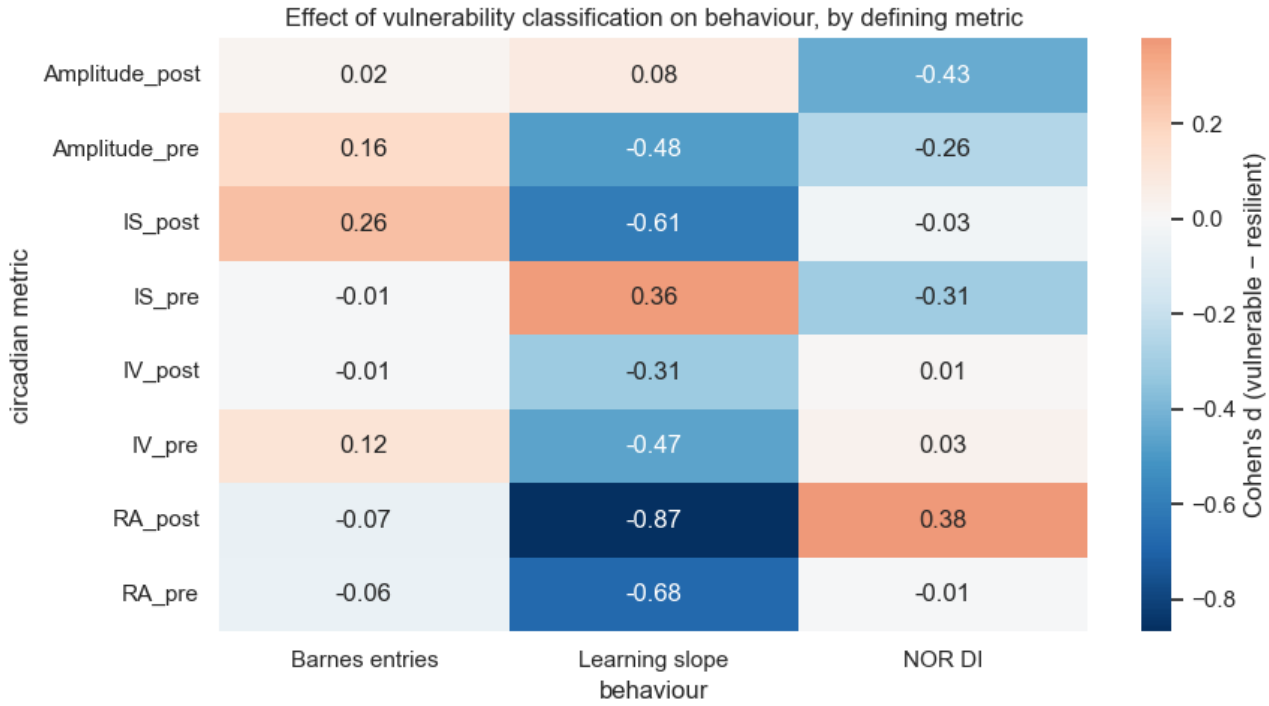
For IS_post (representative), animals in the LOW tail and HIGH tail are behaviourally on opposite sides of the middle on Barnes entries, learning slope, and NOR DI. Pooling both tails into a single 'vulnerable' label destroys the biological signal — core reason the ±1 SD approach fails.

NOTEBOOK 1 — Per-metric labels disagree across metrics



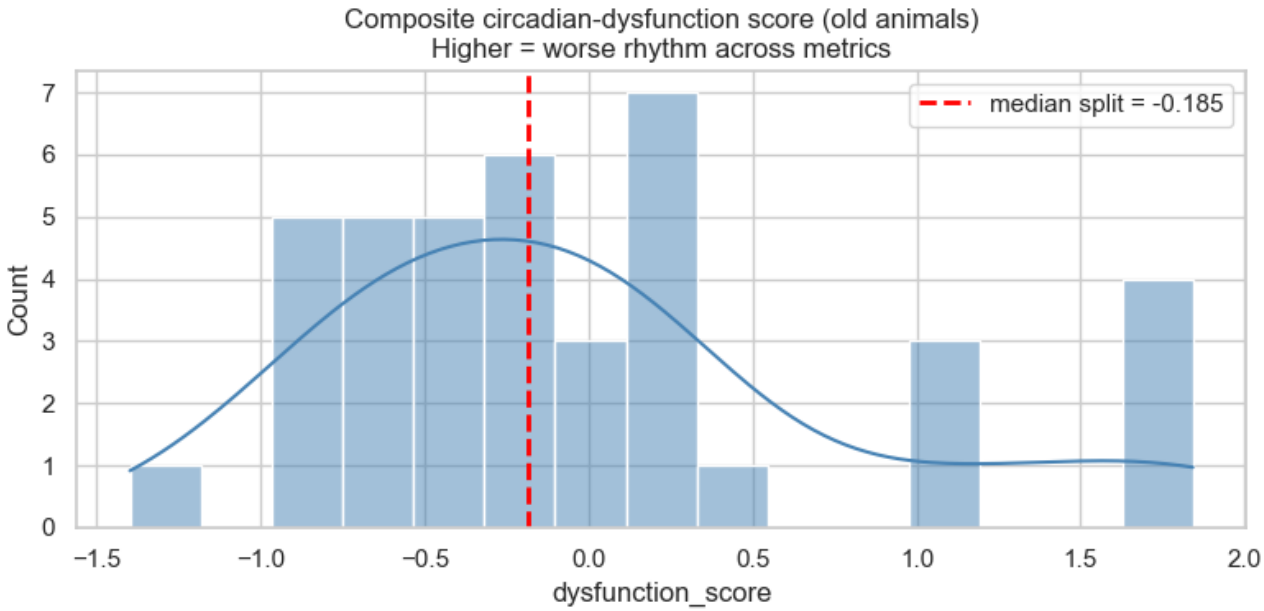
Animal-by-metric heatmap of the vulnerable/resilient label under the PI's ±1 SD rule. The same animal is often labelled differently on different metrics, with agreement well below the 50% chance baseline on many pairs. There is no coherent single classification; which metric you pick changes who counts as 'vulnerable'.

NOTEBOOK 1 — Behavioural conclusions flip with metric choice



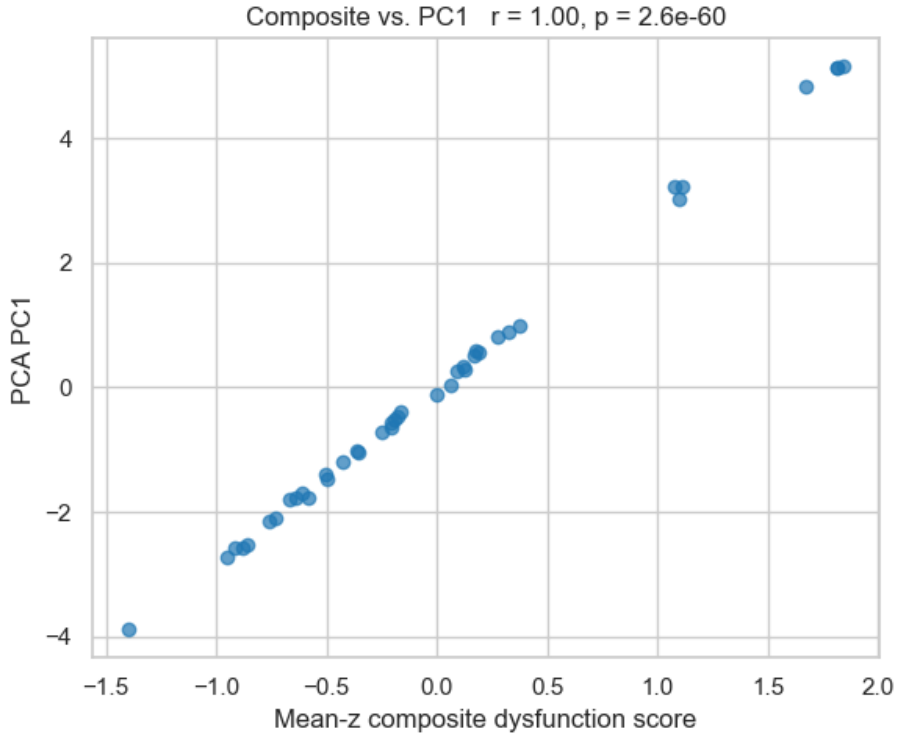
Cohen's d heatmap of (vulnerable – resilient) behavioural differences, one row per circadian metric. Cells are red and blue within the same row — the conclusion 'vulnerables have worse (or better) behaviour' depends entirely on which metric defined the split. This is the researcher-degrees-of-freedom consequence of metric-level classification.

NOTEBOOK 2 — Directional composite dysfunction score (distribution)



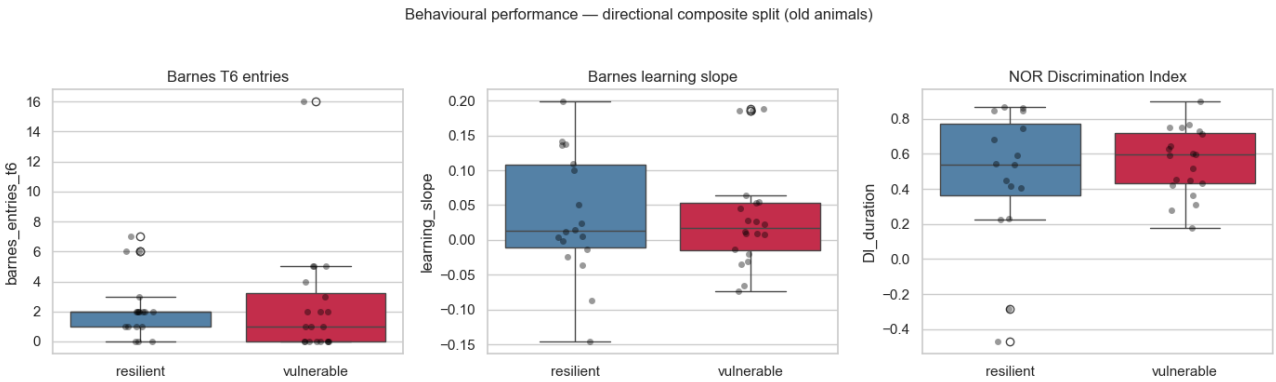
Composite circadian-dysfunction score across old animals, with the median (red) used to split resilient (below) vs vulnerable (above). IS, RA, and Amplitude are sign-flipped so that higher = more dysfunction. The composite fixes four of the six problems from Notebook 1 and sets up a principled single split.

NOTEBOOK 2 — Composite tracks PCA PC1 (internal validation)



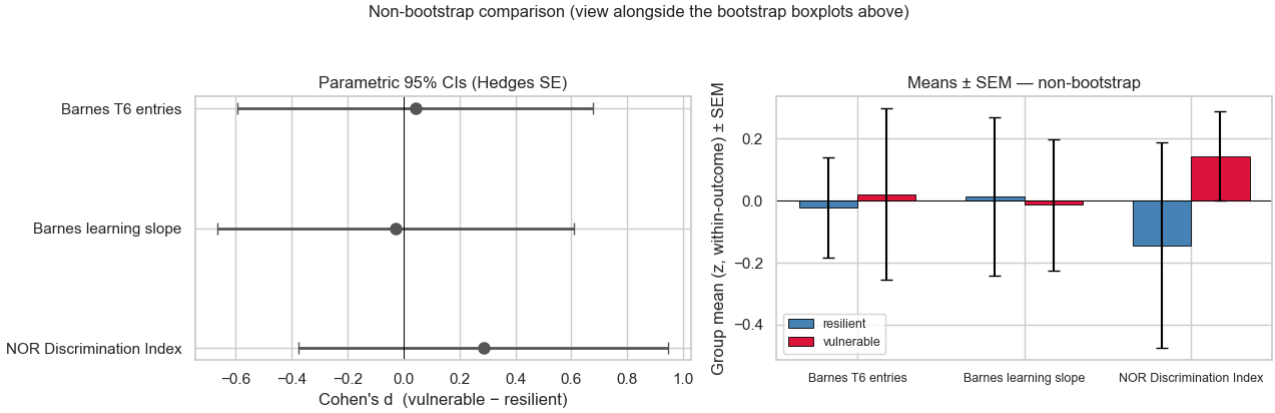
Mean-z directional composite vs the first principal component of the same four circadian metrics. High correlation (typically r > 0.9) shows the composite is capturing the same dominant axis of variation that an unsupervised PCA finds — reassuring that the hand-directed sign flips didn't invent a direction.

NOTEBOOK 2 — Behavioural performance by composite group (null)



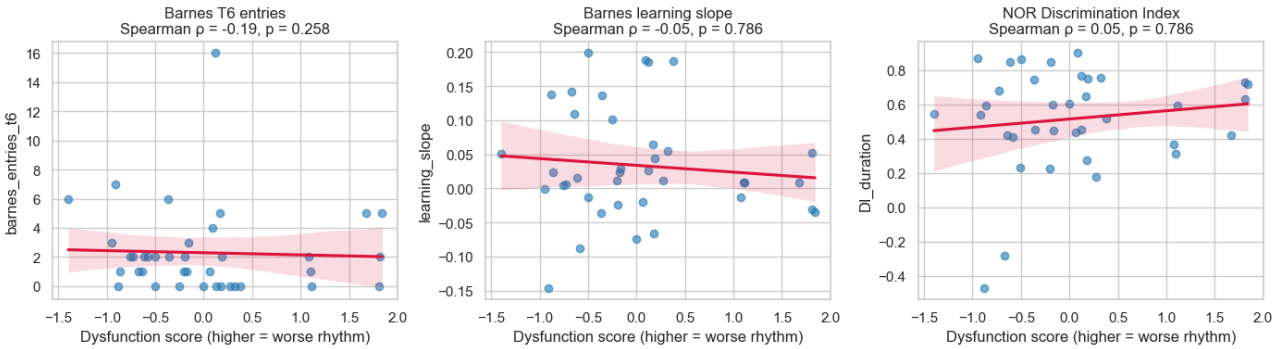
Median-split composite vulnerable vs resilient behavioural comparison. All three outcomes show indistinguishable distributions between groups, consistent with the full-cohort null. Median split in the forward direction does not recover a circadian-cognition relationship.

NOTEBOOK 2 — Non-bootstrap side-by-side validation



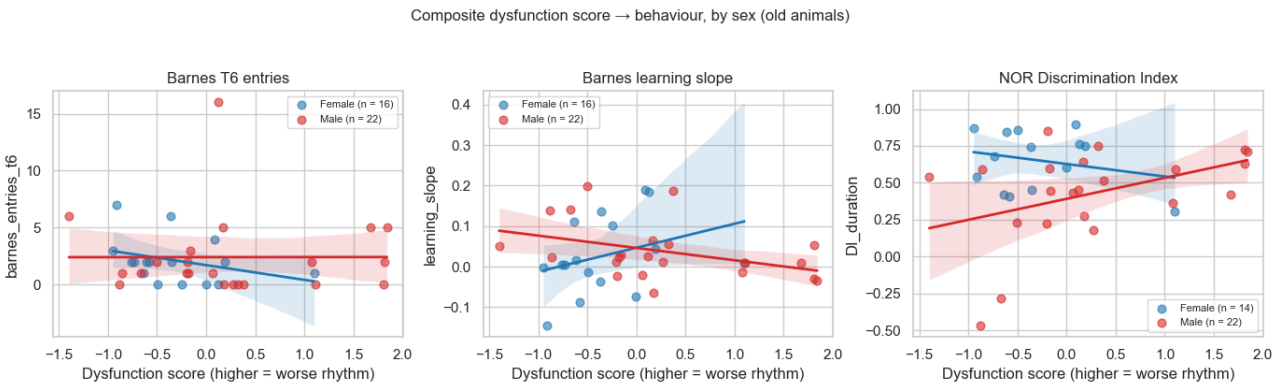
Left: parametric Cohen's d (Hedges SE) forest plot with 95% CIs. Right: group means ± SEM (within-outcome z-scored). Parametric CIs agree with the bootstrap CIs above, confirming the near-zero group differences are not a method artefact. Both approaches cross zero for all three outcomes.

NOTEBOOK 2 — Continuous correlations (full old cohort)



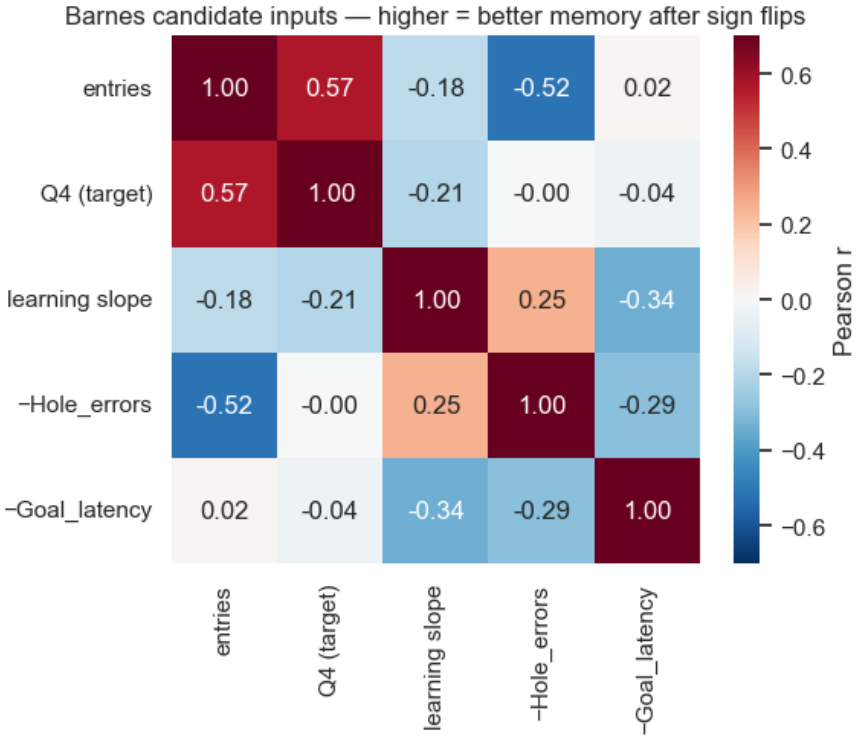
Behavioural outcome vs circadian dysfunction score, continuous form. All three Spearman correlations are near zero and non-significant ($\rho \in [-0.19, +0.05]$, all $p \geq 0.25$). At the pooled old-animal level, neither dichotomisation nor continuous analysis finds a circadian-cognition signal — setting up the need for sex stratification.

NOTEBOOK 2 — Sex-stratified continuous correlations



Within each behavioural outcome, Male (red) and Female (blue) regression lines diverge — especially Barnes learning slope, where Females $\rho = +0.46$ ($p = 0.076$) and Males $\rho = -0.42$ ($p = 0.050$), opposite signs. This sex-dependent pattern is the continuous-form signature of the Old-Male effect that the poster makes explicit.

NOTEBOOK 3 — Barnes input correlations (composite construction)

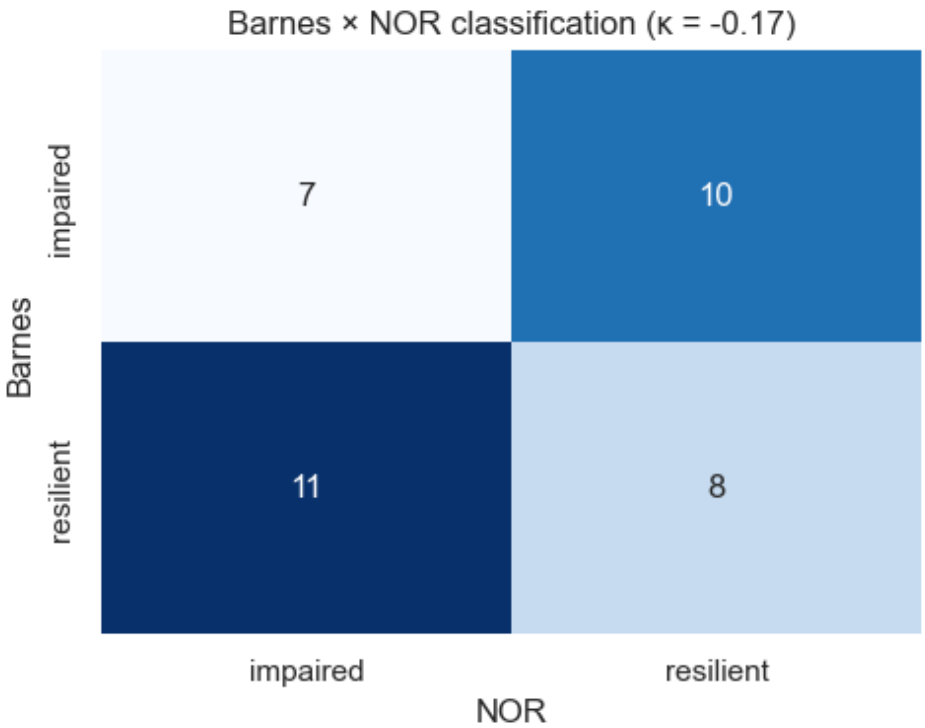


5 × 5 correlation matrix of candidate Barnes inputs (all sign-flipped so higher = better memory). Entries and Q4 co-vary strongly (r = +0.57); Hole_errors is activity-confounded (r = –0.52 with entries); learning_slope and Goal_latency do not belong in the same factor. This diagnostic empirically justifies the final composite (entries + Q4 only).

NOTEBOOK 3 — Barnes fidelity and NOR score distributions

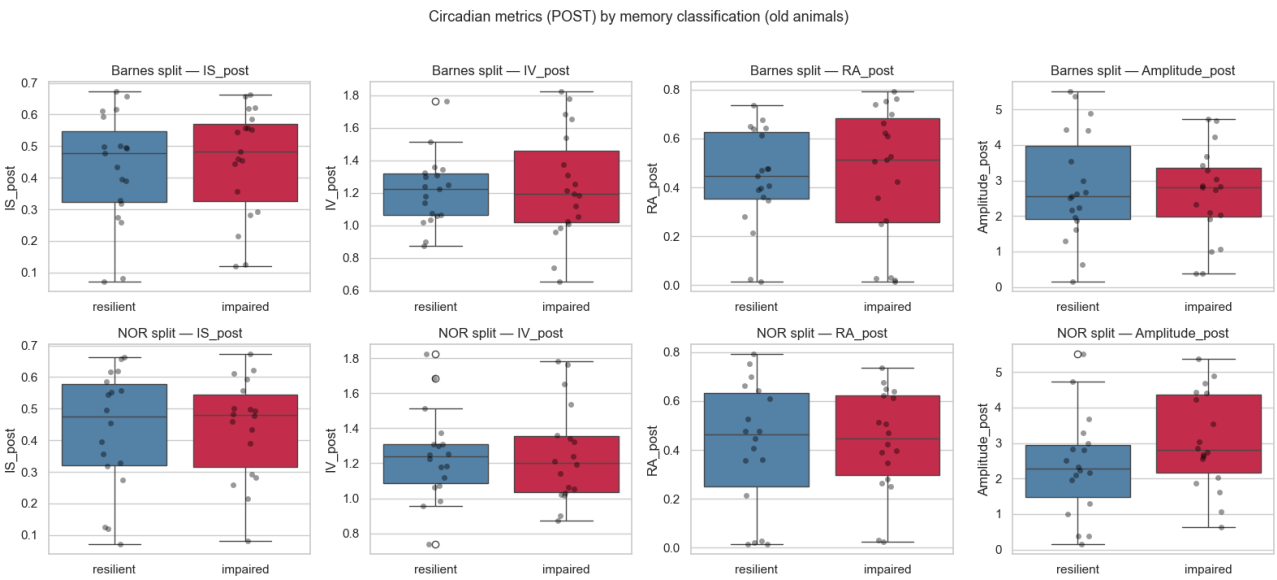
Barnes fidelity composite (z-mean of target entries + target-Q4 %) and NOR discrimination index in old animals, with median splits marked. Both distributions are approximately symmetric around the median, yielding near-balanced 19/19 and 18/18 classifications used in the κ analysis.

NOTEBOOK 3 — Barnes × NOR classification cross-tab



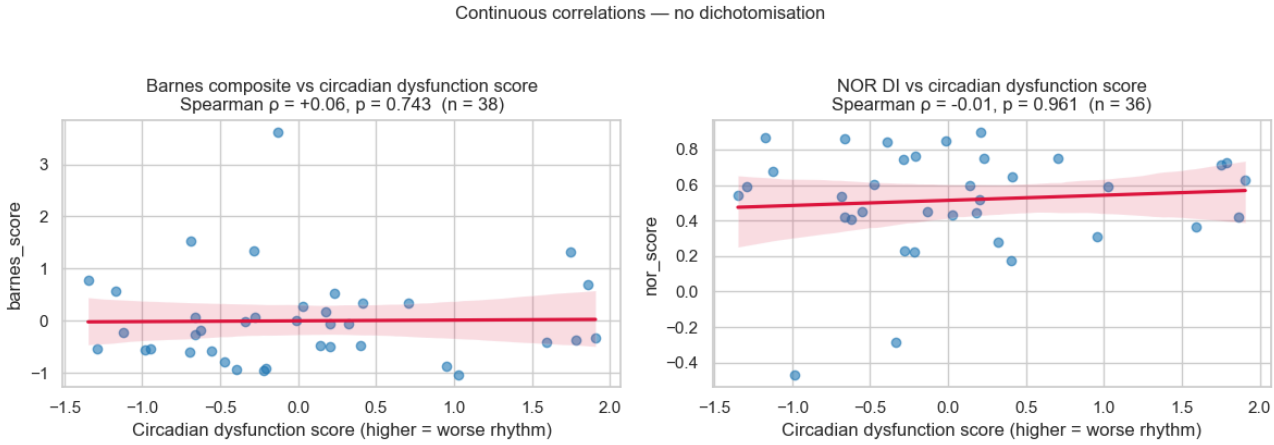
Count of animals in each Barnes × NOR classification cell. Agreement is 41.7% (below the 50% chance baseline). Cohen's κ = –0.17. Memory domains index independent axes of cognitive variation in old animals — the central methodological finding supporting Poster Fig C.

NOTEBOOK 3 — Circadian metrics by memory classification



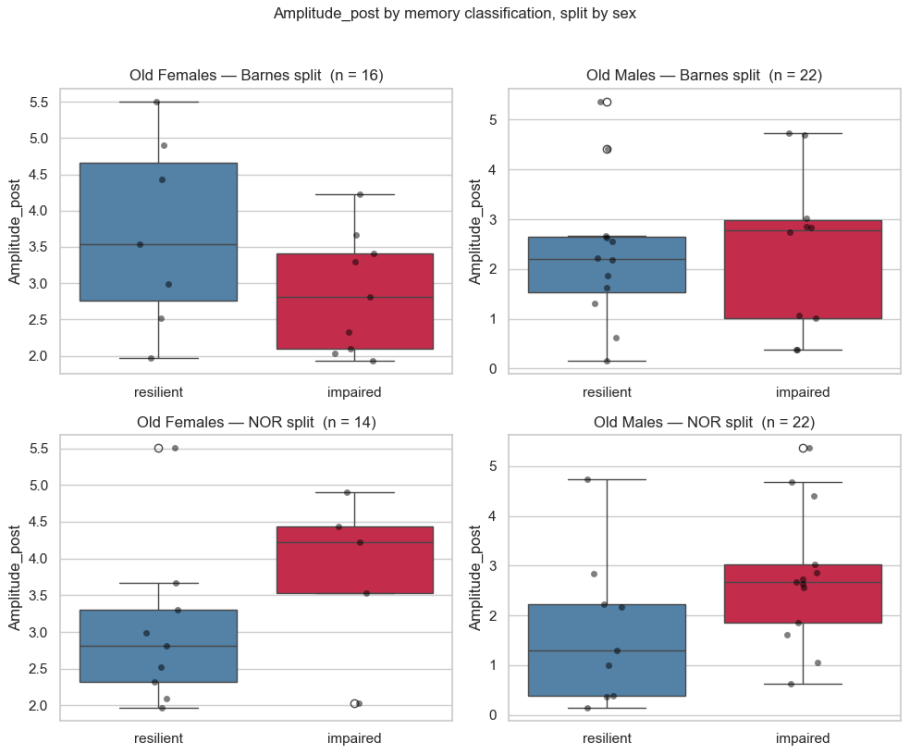
Boxplot grid: each of four circadian metrics by Barnes split (top row) and NOR split (bottom row). Consistent with Fig D in the main poster section, only Amplitude_post by NOR split shows visual separation (NOR-impaired > NOR-resilient, d = +0.51, p = 0.159); all other cells overlap.

NOTEBOOK 3 — Behavioural score vs circadian dysfunction (continuous)



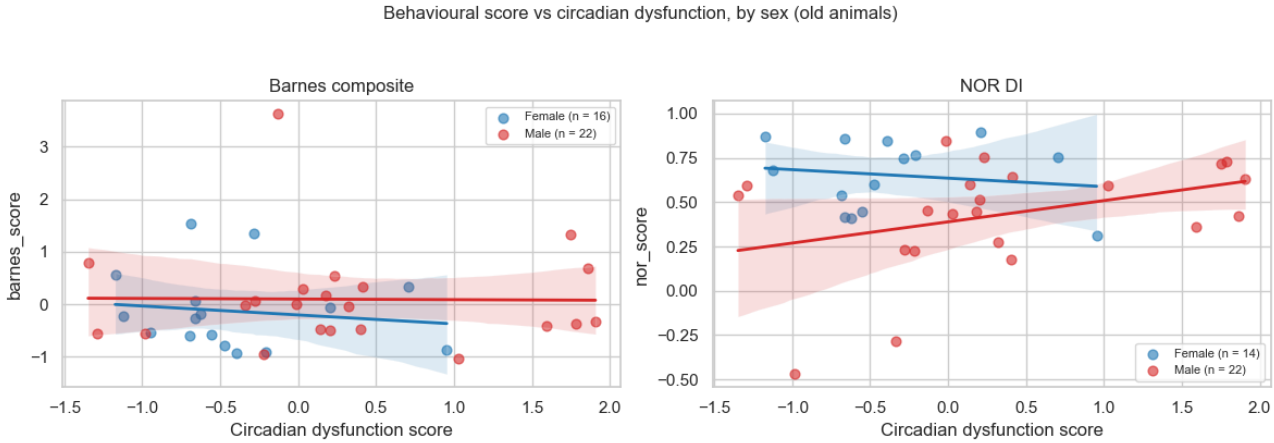
Barnes composite and NOR DI against the circadian dysfunction score, pooled across old animals. Both continuous correlations are near zero, confirming the null at the cohort level before sex stratification. The behaviourally-forward framing recovers the same null as the rhythm-forward framing — two framings, one answer, at the cohort level.

NOTEBOOK 3 — Sex-stratified circadian × memory comparison



Amplitude_post by memory classification (Barnes top, NOR bottom) × Sex. The only visually striking separation is Old-Male × NOR split — replicating the poster's Old-Male Amplitude result from the behaviourally-forward direction.

NOTEBOOK 3 — Sex-stratified continuous correlation



Behavioural score vs circadian dysfunction score, with regression lines by Sex. NOR DI × Males: ρ = +0.35, p = 0.11; Females ρ ≈ 0. Same male-specific NOR–rhythm coupling seen elsewhere, in behaviourally-forward continuous form.

Caveats (must appear on the poster)

1. Old-Male findings are exploratory. They do not survive strict FDR across the 48-test full-cohort family (min p_fdr ≈ 0.93) or the 16-test Old-Male grid (min p_fdr ≈ 0.20). The evidence is a coherent pattern across multiple related metrics, not a single corrected test.

2. Small subgroup n. Old-Male total n = 22; strongest single result (ISF cell) n = 10. Bootstrap CI [–0.78, –0.10] is wide; replication on an independent cohort is required.

3. Paradoxical direction needs a mechanism. Stronger rhythm → worse NOR in aged males is opposite to the naive "rhythm loss impairs cognition" model. Candidate frames: rigid rhythms reduce the cognitive flexibility needed for novelty processing; compensatory amplitude amplification in homeostatically-compromised animals. Present this as hypothesis-generating, not confirmed.

4. Intervention did not work as advertised. ISF does not differentially alter rhythm metrics (Section 1). Do not write "ISF-induced circadian disruption" anywhere — ISF appears to moderate rather than produce the phenotype.

5. Memory domains are orthogonal. Barnes and NOR cannot be collapsed into a single memory score ($\kappa = -0.17$). Report them separately, always.

For the full statistical methods and an expanded narrative, see *poster_interpretation.txt* in the same folder.